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|----------------------------------------------------------------------------------|--------------------------------------------------------------|----------------|--------------------|
|  | Work Instruction                                             | Doc. Revision  | <b>00</b>          |
|                                                                                  | <b>AXI V810 STD Y-Axis Glass Scale Replacement Procedure</b> | Effective Date | <b>25 Jan 2017</b> |

**This procedure is applicable to below models:**

|           |                                     |    |                                     |    |
|-----------|-------------------------------------|----|-------------------------------------|----|
| V810 STD  | <input checked="" type="checkbox"/> | S1 | <input checked="" type="checkbox"/> | S2 |
| V810 XXL  | <input type="checkbox"/>            | S1 | <input type="checkbox"/>            | S2 |
| V810 S2EX | <input type="checkbox"/>            |    |                                     |    |
| V810 Mini | <input type="checkbox"/>            |    |                                     |    |

#### **REVISION HISTORY**

| Rev | Effective Date     | Description   | Doc Originator |
|-----|--------------------|---------------|----------------|
| 00  | <b>25 Jan 2017</b> | First Release | TAN LIANG KAI  |
|     |                    |               |                |
|     |                    |               |                |

|                                                                                  |                                                              |                |                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------|----------------|--------------------|
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## Content

|                                                |   |
|------------------------------------------------|---|
| A. Y-Axis Glass Scale.....                     | 3 |
| I. Y-Axis glass scale removal.....             | 3 |
| II. Replacement of the Y-Axis Glass Scale..... | 5 |

|                                                                                  |                                                       |                |             |
|----------------------------------------------------------------------------------|-------------------------------------------------------|----------------|-------------|
|  | Work Instruction                                      | Doc. Revision  | 00          |
|                                                                                  | AXI V810 STD Y-Axis Glass Scale Replacement Procedure | Effective Date | 25 Jan 2017 |

## A. Y-Axis Glass Scale

### I. Y-Axis glass scale removal

Y-Axis glass scale assembly Part ID: 4XASP-A121

1. Turn off x-ray source and open front access door.
2. Remove 4 screws and gentle remove Y-Axis cover plate as shown in Figure 1.



Figure 1

3. Remove spring clamps ,home sensor & limit sensor as shown in Figure 2.

**Remark: Perform minor marking on current limit sensor & home sensor position for ease of adjustment**

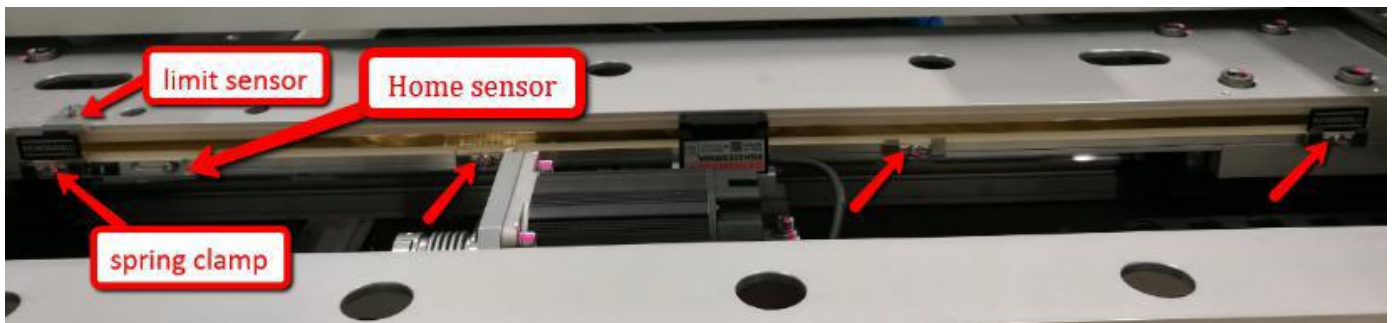
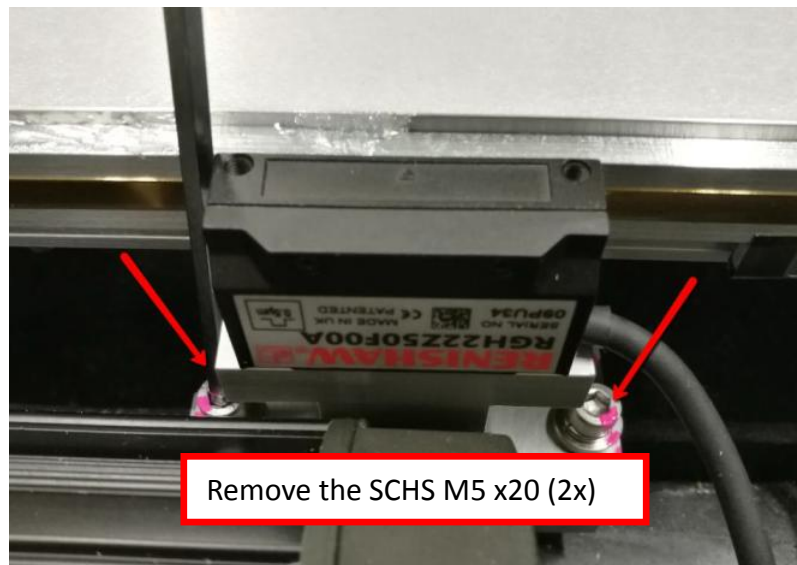


Figure 2 Glass scale

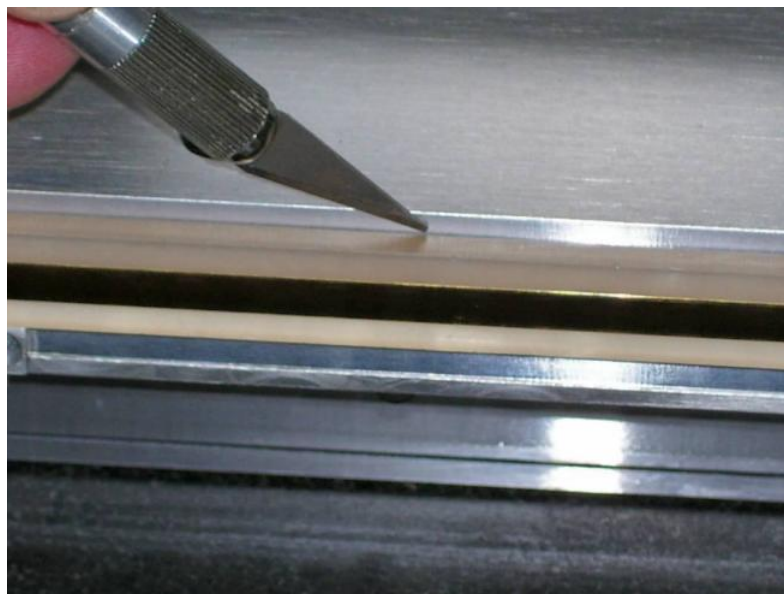
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|----------------------------------------------------------------------------------|-------------------------------------------------------|----------------|-------------|
|  | Work Instruction                                      | Doc. Revision  | 00          |
|                                                                                  | AXI V810 STD Y-Axis Glass Scale Replacement Procedure | Effective Date | 25 Jan 2017 |

4. Gently remove Y axis encoder as below Figure 3 and allocate at visible position.



*Figure 3 Y axis encoder*

5. Cut the RTV glue which secure glass scale.



*Figure 4 RTV glue removal*

6. Gentle remove Y-Axis glass scale by pulling it in the direction as shown in Figure 5.



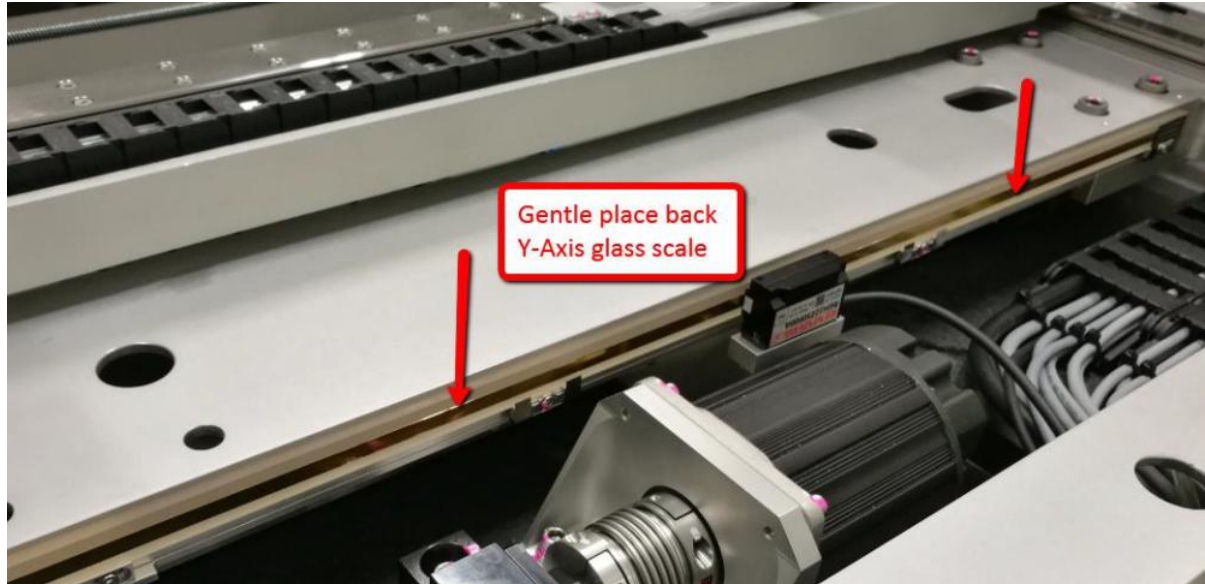
*Figure 5 Glass scale removal*

|                                                                                  |                                                       |                |             |
|----------------------------------------------------------------------------------|-------------------------------------------------------|----------------|-------------|
|  | Work Instruction                                      | Doc. Revision  | 00          |
|                                                                                  | AXI V810 STD Y-Axis Glass Scale Replacement Procedure | Effective Date | 25 Jan 2017 |

## II. Replacement of the Y-Axis Glass Scale

1. Gentle assemble new Y-Axis glass scale as shown in Figure 6.

**Remark: Caution on glass scale orientation.**



*Figure 6 Glass scale assembly*

2. Position the glass scale lining up the termination tag of the glass scale with the spring clamp.



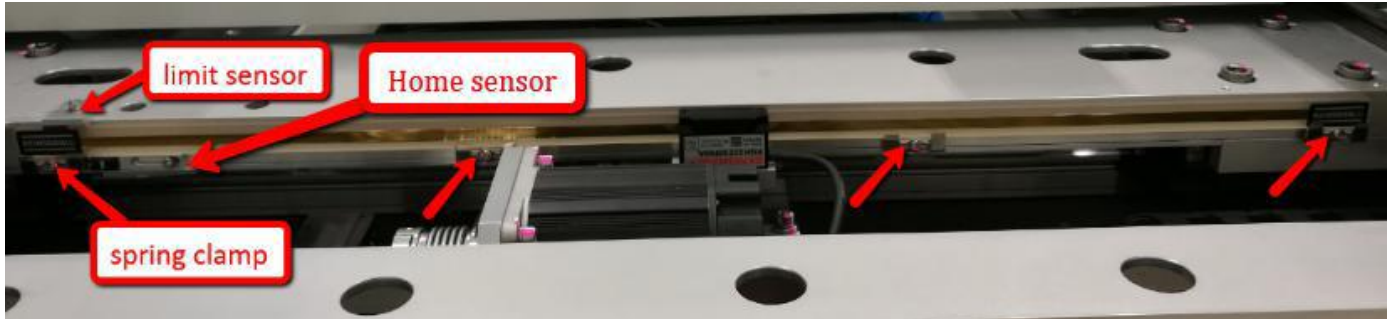
*Figure 7 Glass scale lining up (Reference Only)*



|                                                                                  |                                                       |                |             |
|----------------------------------------------------------------------------------|-------------------------------------------------------|----------------|-------------|
|  | Work Instruction                                      | Doc. Revision  | 00          |
|                                                                                  | AXI V810 STD Y-Axis Glass Scale Replacement Procedure | Effective Date | 25 Jan 2017 |

- Tighten all screws which secure spring clamps as shown in Figure 8.

**Remark: Make sure spring clamps are well-positioned and do not contain any friction with encoder while moving the Y-Axis stage.**



*Figure 8 Glass scale replacement*

- Assemble Y axis encoder as below Figure 9.



*Figure 9 Y axis encoder assembly*

- Assemble Y-Axis cover plate and tighten 4 x screws as shown in Figure 10.



*Figure 10 Y axis cover plate*

|                                                                                  |                                                              |                |                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------|----------------|--------------------|
|  | Work Instruction                                             | Doc. Revision  | 00                 |
|                                                                                  | <b>AXI V810 STD Y-Axis Glass Scale Replacement Procedure</b> | Effective Date | <b>25 Jan 2017</b> |

- Move X-Axis stage to minimum hard stop (Front side) and maximum hard stop (Rear side) . Visual check Y-Axis encoder which require green light Illuminated along the way as shown in Figure 11.  
If adjustment require,may refer **“AXI V810 Y-Axis Read Head Encoder Replacement Procedure ”**

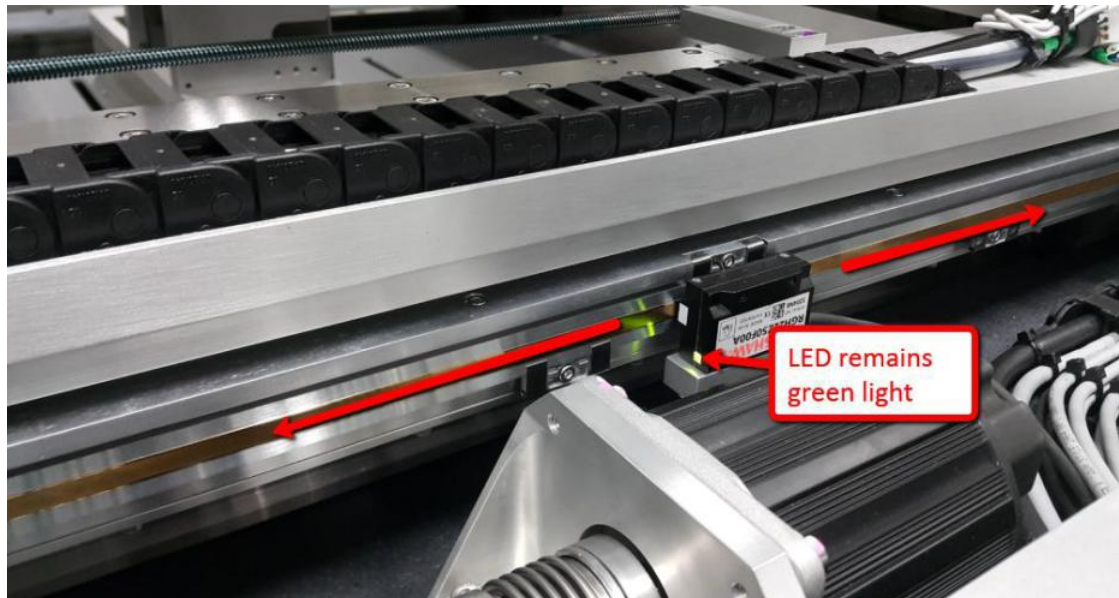


Figure 11 Y axis encoder checking

- Assemble limit sensor and home sensor by refer to origin location(By refer to Step 3).
- Close front access door
- Open V810 > Services > Subsystem Status & Control > Panel Positioner > Initialize

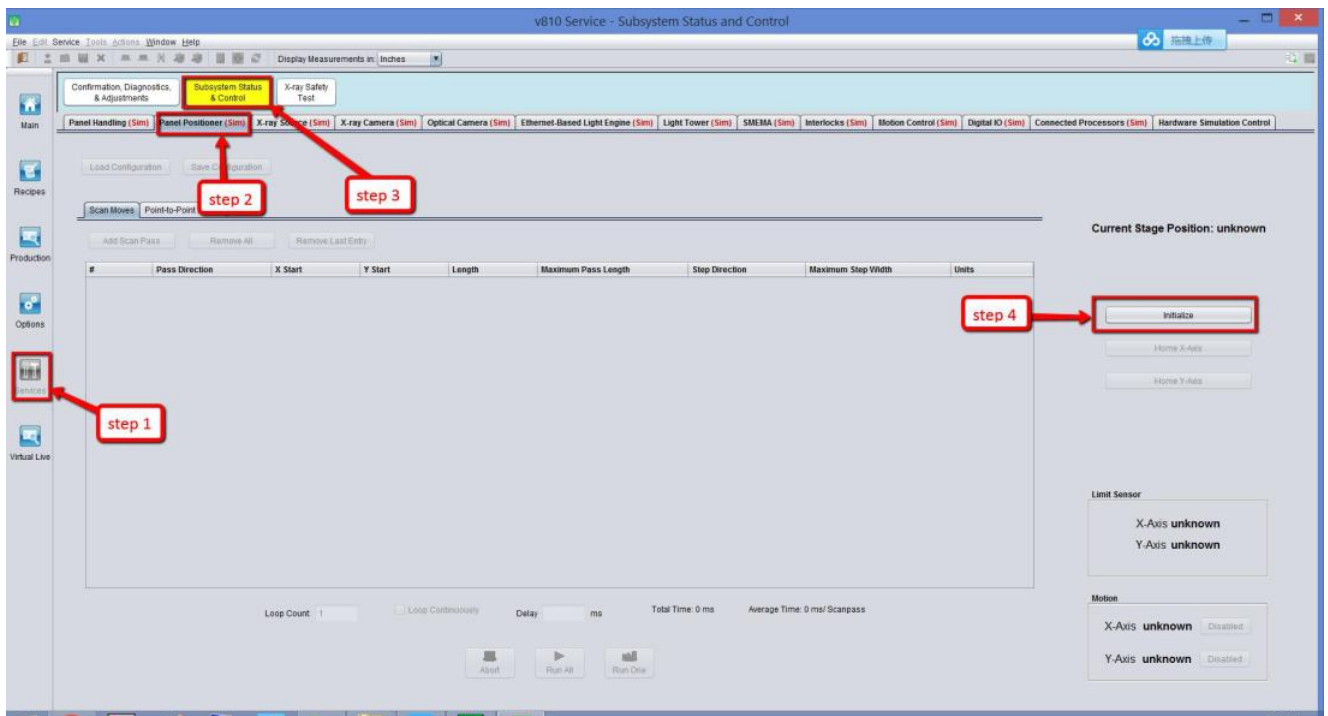


Figure 12 Panel Positioner

|                                                                                  |                                                              |                |                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------|----------------|--------------------|
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10. Open front access door.
11. Disable Y-Axis motion at Panel Positioner & perform below checking.
  - a) DO not move XY Stage and may check Limit sensor status at Panel Positioner where must stated as “Clear” during home position

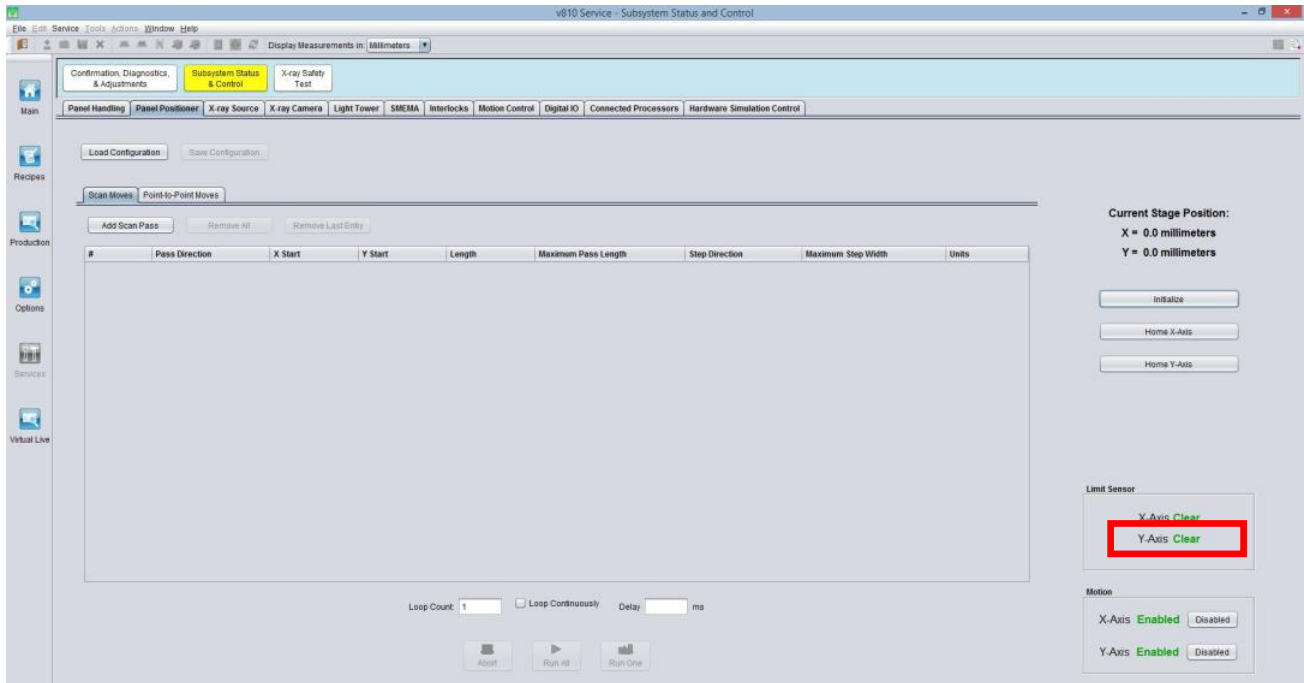


Figure 13 Y axis Limit Sensor (CLEAR)

- b) Move Y axis stage towards minimum hard stop (Front side) & check Limit sensor where must stated as “Blocked”.

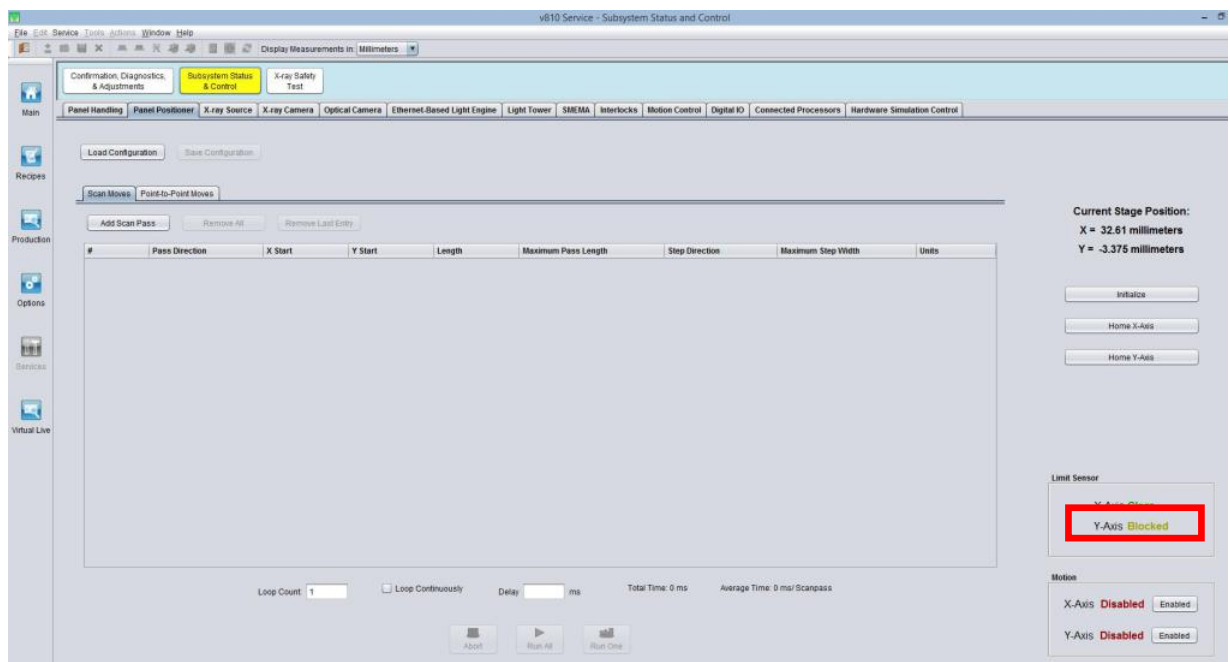


Figure 14 X axis Limit Sensor (BLOCKED)

**Remark: If result is not meet to Step 11a ~11b requirement ,may refer to “AXI V810 STD Y-Axis Limit Sensor Adjustment Procedure” for further adjustment.**



|                                                                                  |                                                       |                |             |
|----------------------------------------------------------------------------------|-------------------------------------------------------|----------------|-------------|
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|                                                                                  | AXI V810 STD Y-Axis Glass Scale Replacement Procedure | Effective Date | 25 Jan 2017 |

- c) Move Y axis stage towards minimum hard stop(Front side).
- d) Check the value for the Y-axis in current stage position must within 3.00mm ~ 3.30 mm.



Figure 15 Y axis (Current Stage Position)

**Remark: If result is not meet to Step 11c ~11d requirement ,may refer to “AXI V810 STD Y-Axis Home Sensor Adjustment Procedure” for further adjustment.**

12. Run 3 loops of long scan and 1 loop of CD &A.